Chapter 3b - Remote Logins

Remote Login

- Remote login is a way to remotely access and control a computer or device, typically over a network connection. It allows a user to log in to the remote system, perform tasks, and access resources as if they were physically present at the machine. Remote login is often used for remote access and management of servers, networking devices, and other equipment that need to be accessed and maintained remotely.

- It is a useful tool for system administrators, network engineers, and other IT professionals who need to perform tasks on multiple systems or devices remotely..

- So people can work from home from the bed

- TELNET - simple, fast, not secure
- SSH - command only, secure connection
- RDP - remote desktop protocol (Windows)
- VNC
- ZOOM
- MS TEAM

TELNET

TELNET

- Telnet is a network protocol that allows a user to remotely access and control a device over a network connection. It was one of the first protocols developed for remote login, and it is still in use today for some applications.

- When using Telnet, a user connects to a remote system using a Telnet client program, which sends commands and receives output from the remote system over a network connection. Telnet allows the user to log in to the remote system using a username and password, and perform tasks and access resources as if they were physically present at the machine.

- Telnet is typically used for remote access and management of servers, network devices, and other types of equipment. It is a simple and widely supported protocol, but it has some security vulnerabilities, as it does not encrypt the data transmitted over the network connection. For this reason, Telnet is not recommended for use in sensitive environments or for transmitting sensitive data.

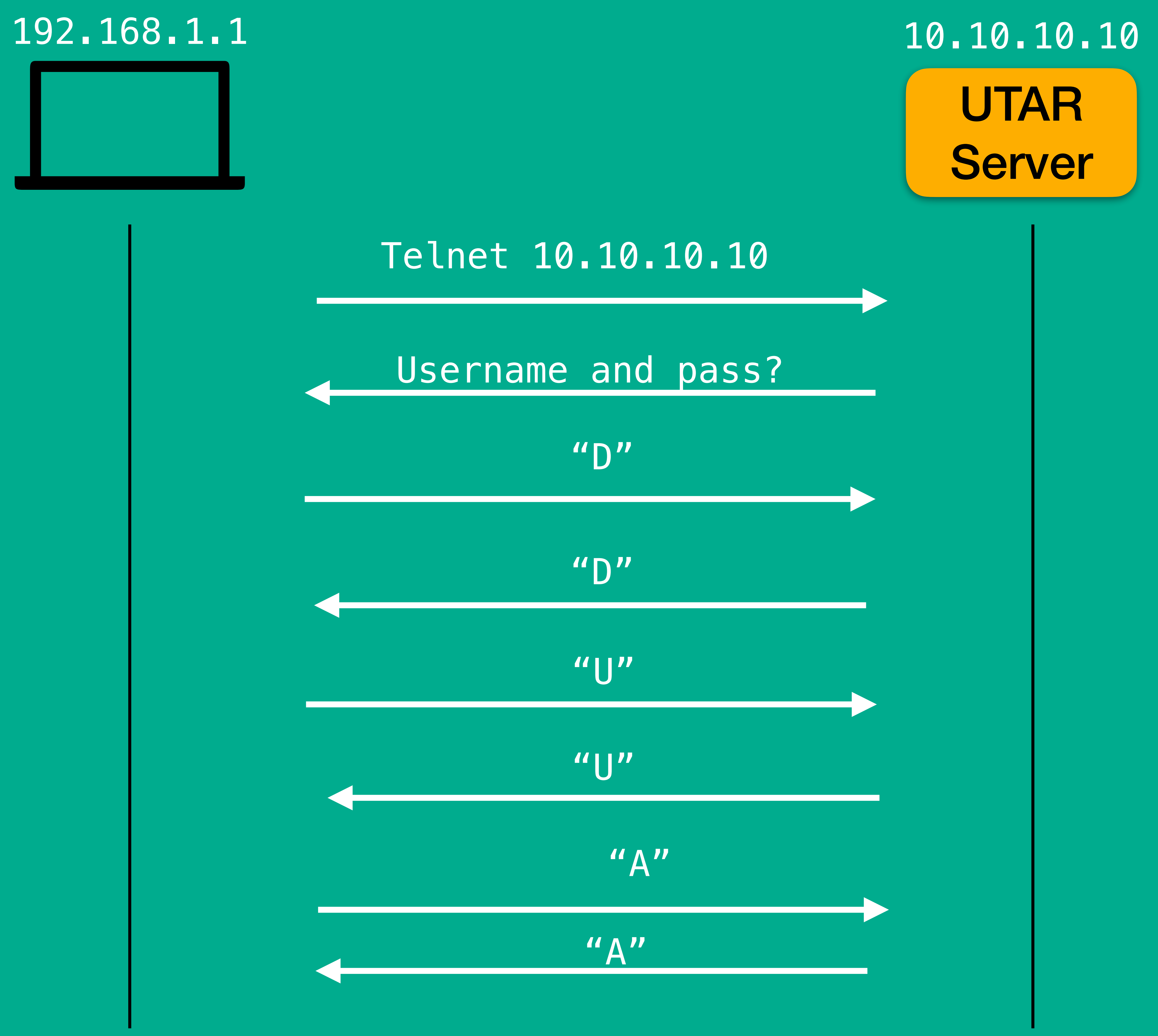
TELNET DEMO

(Packet Tracer)

TELNET

- Send usernames and password in plaintext
- Send 1 character at a time

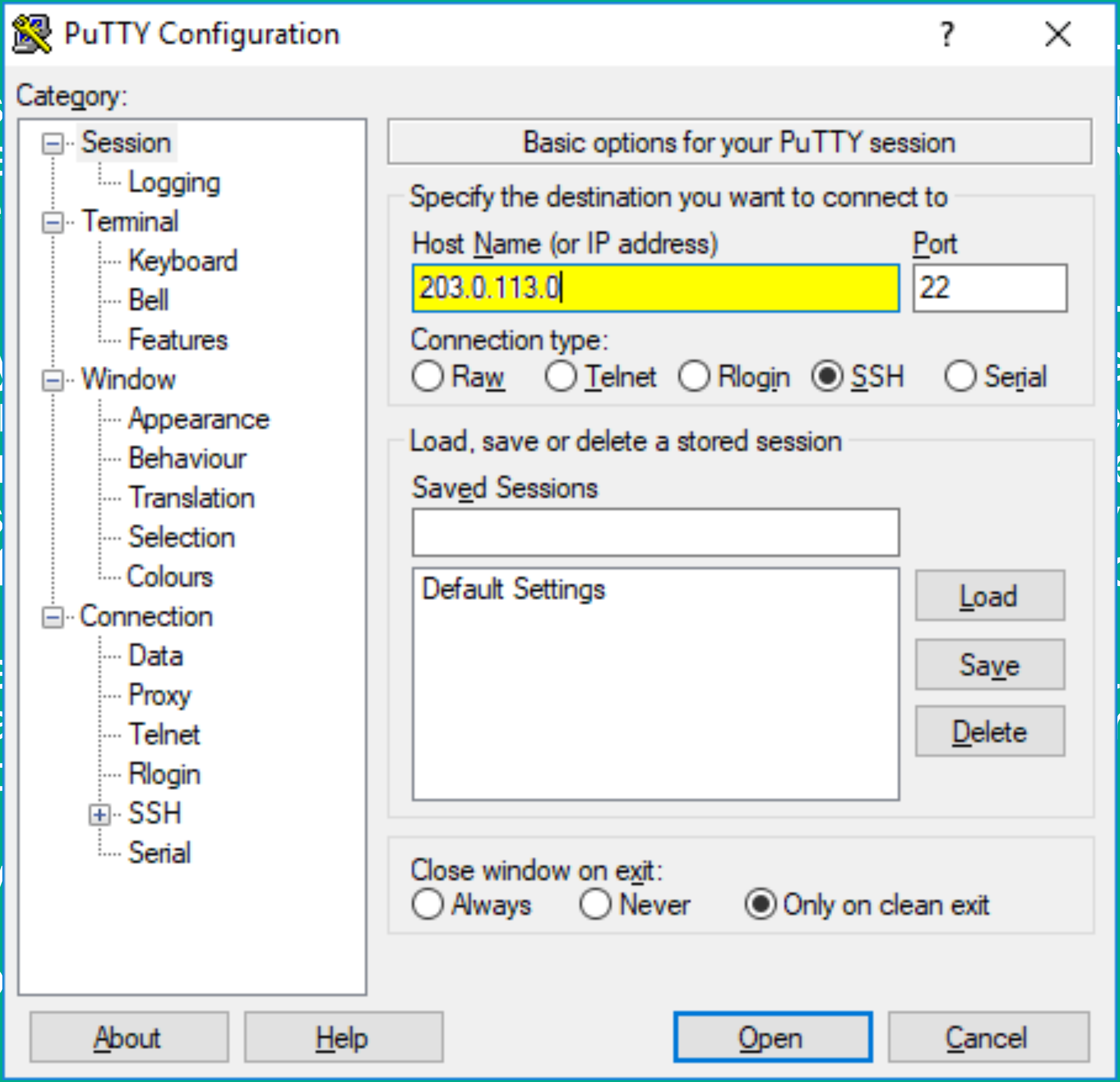
- For example,
    - Username: avo
    - Password: dua



TELNET packet dumps

Secure Shell (SSH)

SSH

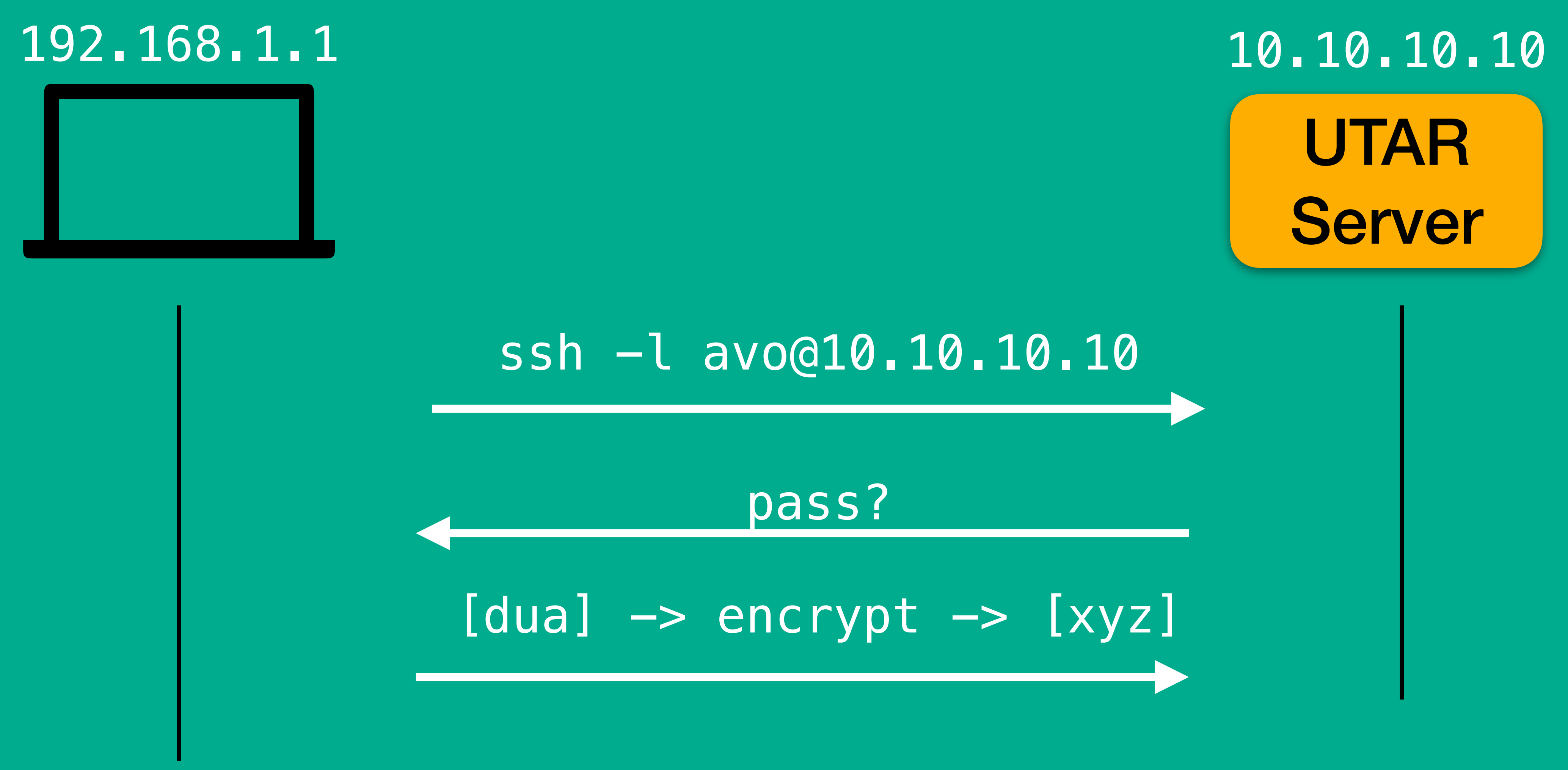


- **SSH (Secure Shell)** … to a remote system and
  execute commands … nt for Telnet and other
  insecure remote she … n and encrypted
  communication ove
- When using SSH, a … H client program, which
  establishes an encry … er can then authenticate
  using a username ar … y pair. Once
  authenticated, the u … em and access resources
  as if they were phys … used for remote access
  and management of … quipment.
- It is a flexible and se … ny platforms. SSH can
  also be used to esta … rding network
  connections, and ot
- There are a few way
  - putty (ssh client)
  - command promp

SSH

- Send usernames and password in cipher text
- Send the whole user/pass credential data block together

- For example,
  - Username: avo
  - Password: dua



SSH DEMO

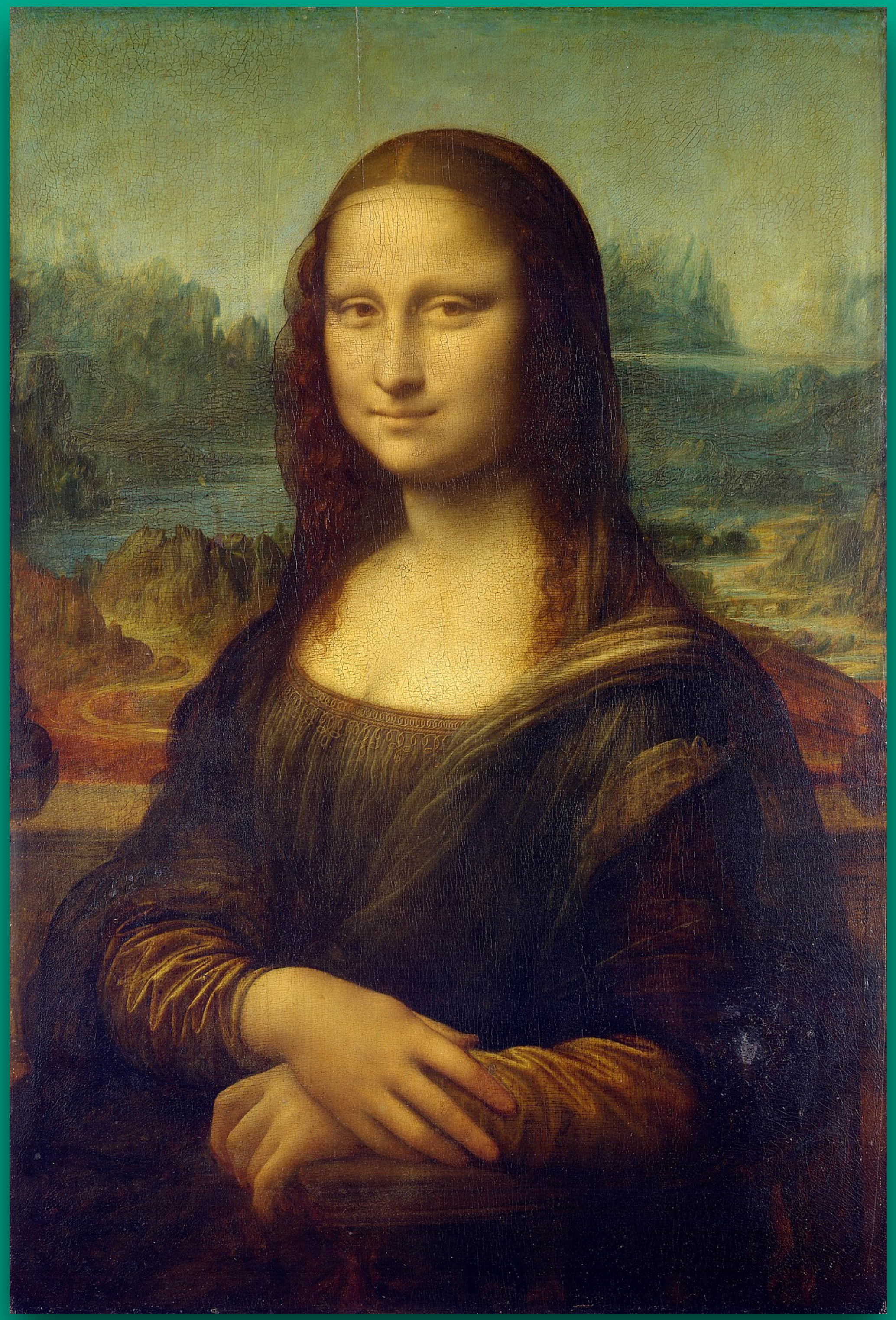
(Packet Tracer)

TELNET vs SSH

Encryption in a nutshell

Encryption is the process of converting plaintext (readable text) into ciphertext (unreadable text) using a key. The key is a secret value that is used to encrypt and decrypt the data. The process of encrypting the data makes it unreadable to anyone who does not have the key, thus ensuring the confidentiality and security of the data.

POV of owners

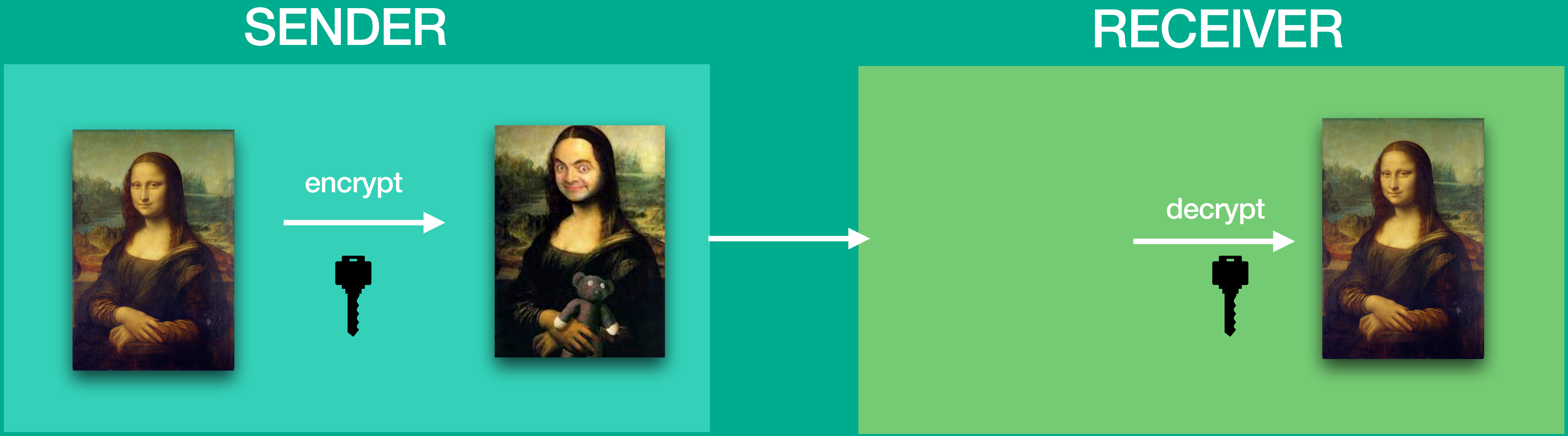


encrypt →

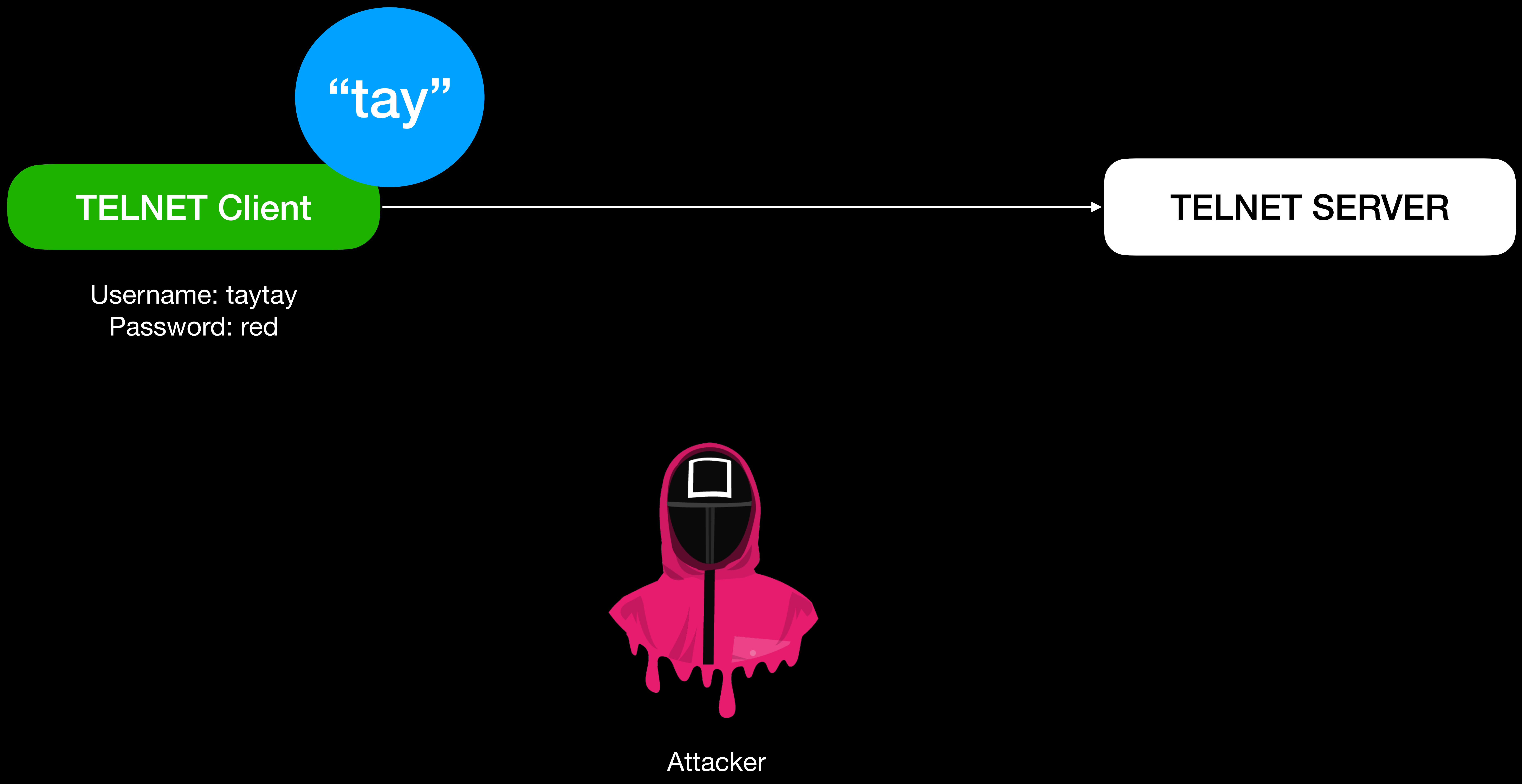
← decrypt

POV of others

How encryption works?

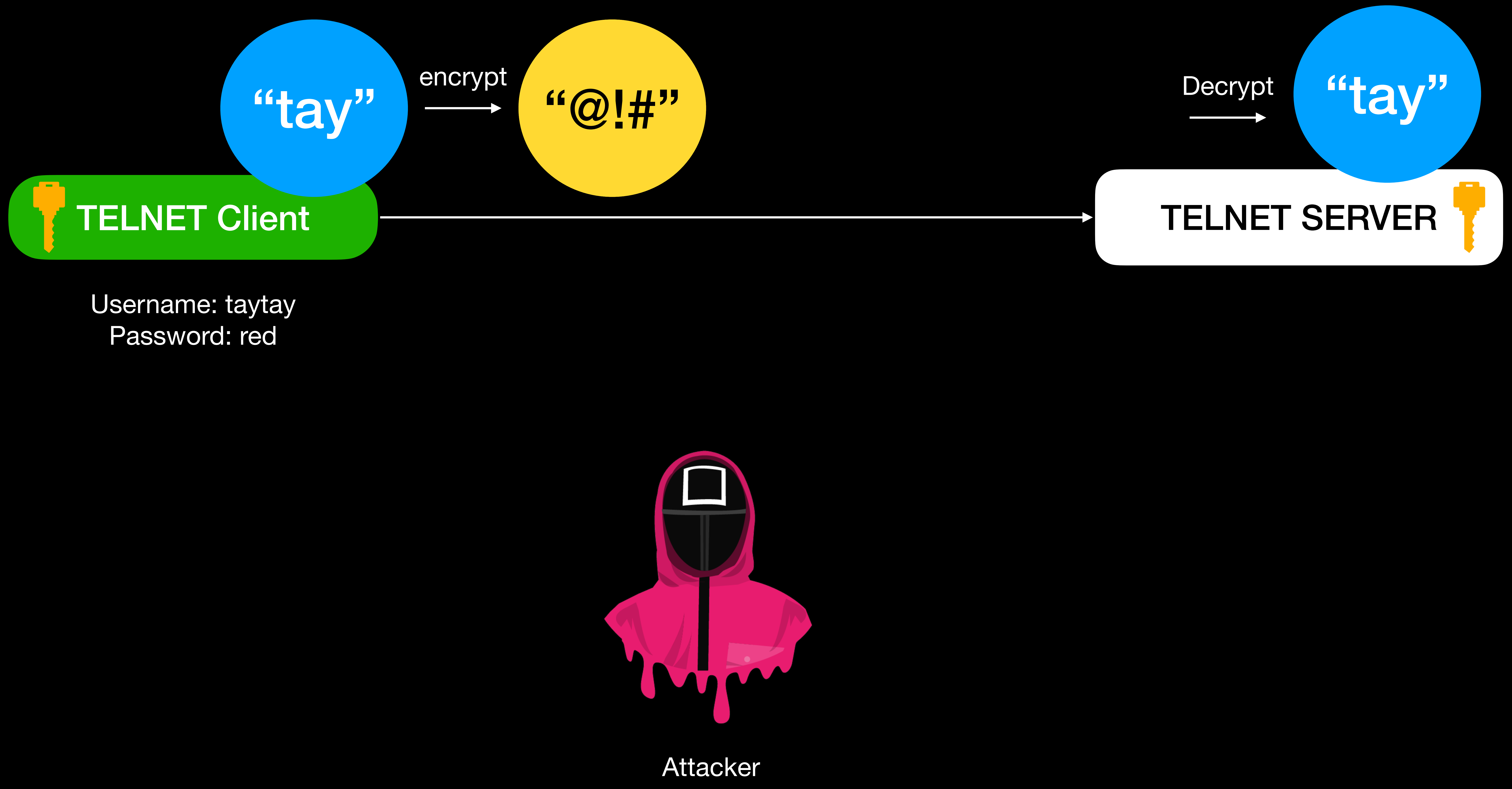


TELNET is vulnerable to MITM



SSH encrypts plaintext into cipher text before sending the data.

When MITM intercepts this packets, the hackers only get the ciphertext. Since the hacker do not have the key used to encrypt the plaintext, they cannot decrypt the ciphertext into the original data



TELNET vs SSH

- When to use which protocol?

Key	Telnet	SSH
Definition	Telnet is the joint abbreviation of Telecommunications and Networks and it is a networking protocol best known for UNIX platform designed specifically for local area networks.	SSH or Secure Shell is a program to log into another computer over a network, to execute commands in a remote machine, and to move files from one machine to another.
Operation	Telnet uses the port 23 and it was designed specifically for local area networks.	SSH on other hand runs on port 22 by default however it can be easily changed.
Security	As compared to SSH, Telnet is less secured.	SSH is a very secure protocol because it shares and sends the information in encrypted form
Data format	Telnet transfers the data in simple plain text.	SSH uses Encrypted format to send data and also uses a secure channel.
Authentication	No authentication or privileges are provided for user's authentication.	SSH is more secure, so it uses public key encryption for authentication.
Preference	Due to its less security provisions, Telnet is recommended only for private networks.	SSH is suitable for Public networks.

VNC (Virtual Network Computing)

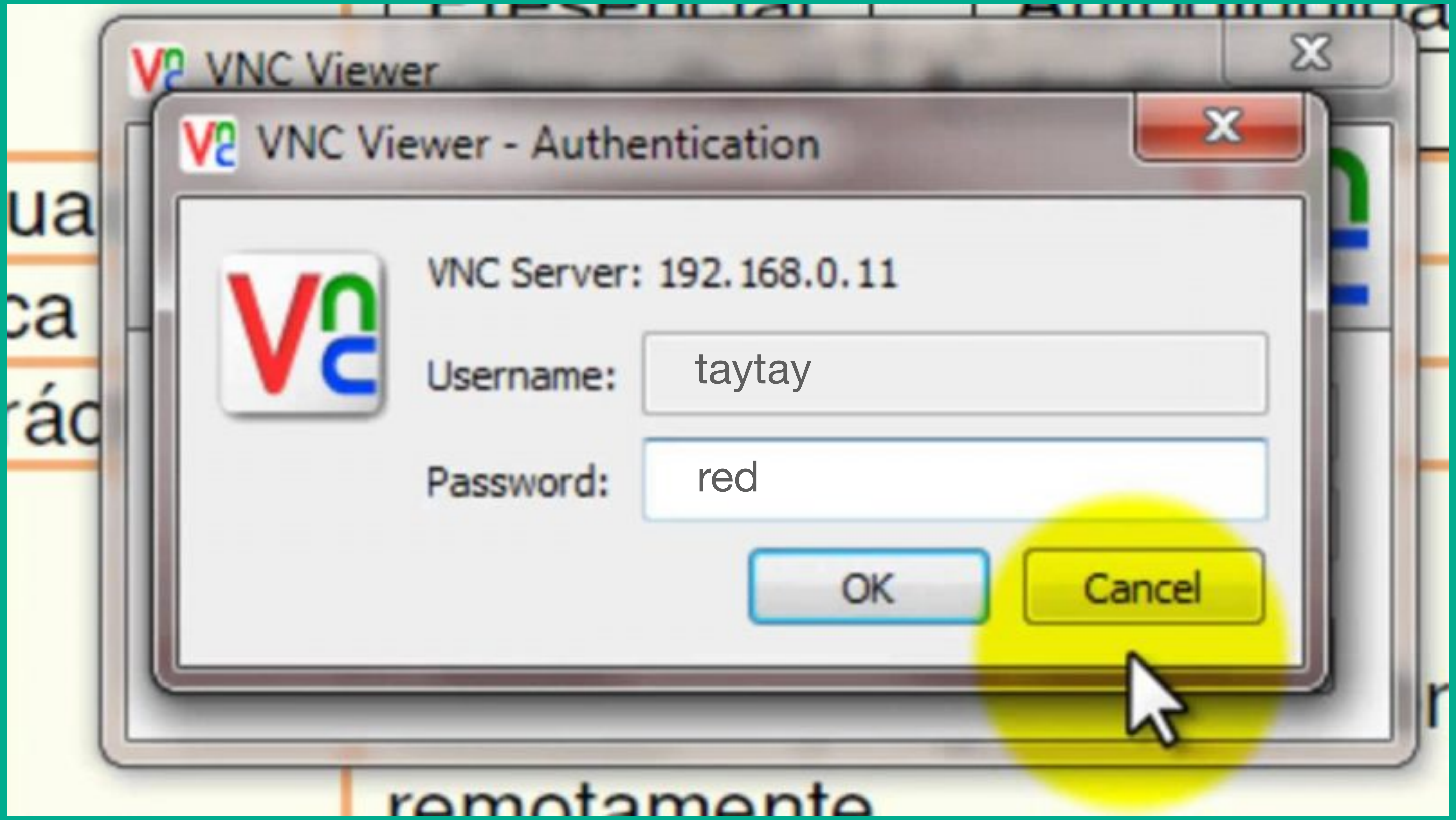
VNC

- **VNC (Virtual Network Computing)** is a remote control software that allows you to view and interact with the desktop of one computer (the "server") using another computer or mobile device (the "client"). VNC uses a client-server architecture, where the VNC server software runs on the computer that you want to control, and the VNC client software runs on the computer or device that you will use to remotely access the server. The VNC server and client communicate over a network connection, allowing you to view and control the server's desktop from any location. The service VNC refers to is the background daemon that run the vnc protocol allowing these communications and control to happen.

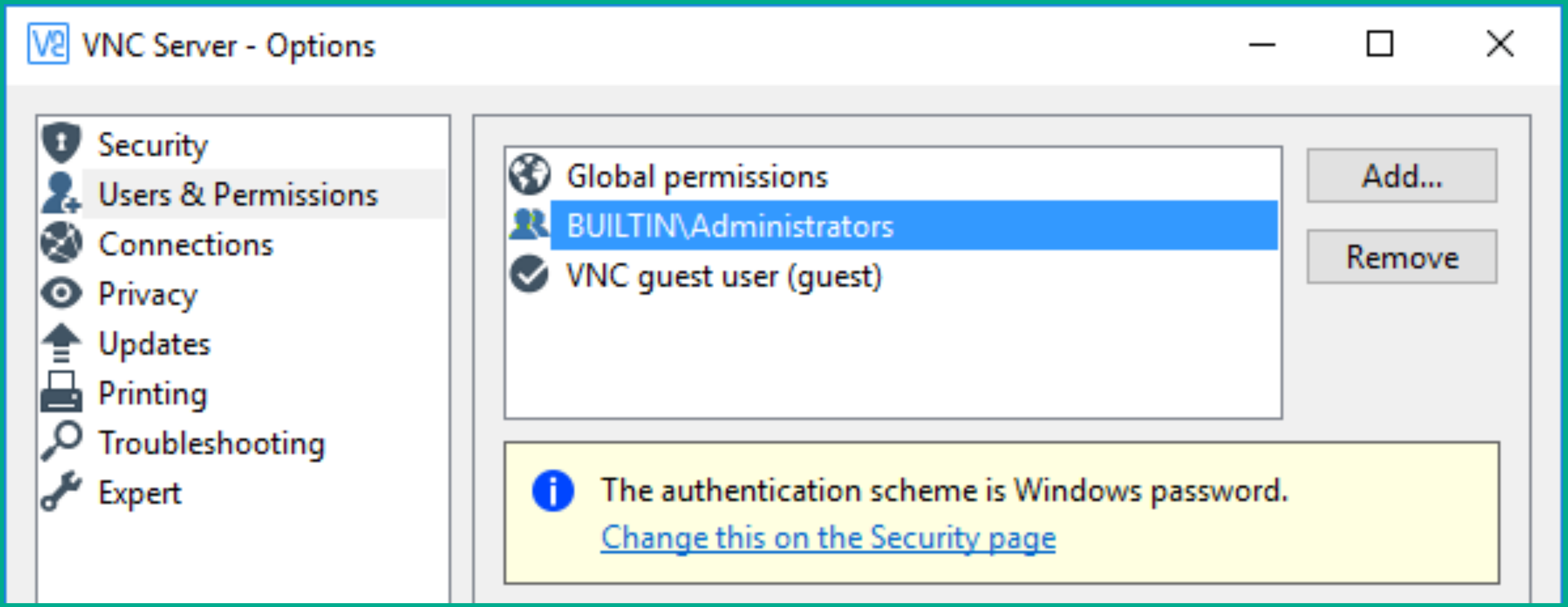
- Think of TELNET/SSH but with UI.

VNC

- **VNC CLIENT**



VNC SERVER



Username: taytay
Password: red

The PC you use to work

The PC you want to connect into
(to get your data)

Security Recommendations

For remote logins

Setting strong passwords

PasswordMonster

info@passwordmonster.com

How Secure is Your Password?

Take the Password Test

Tip: Avoid sequences or repeated characters in your passwords

Show password: ☑

Type a password

No Password

0 characters containing: Lower case Upper case Numbers Symbols

Time to crack your password:

0 seconds

Dictionary Attacks

Try most probable combinations of characters and symbols using heuristics

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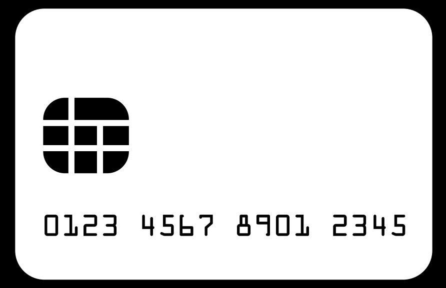
Try most probable combinations of characters and symbols using heuristics



Instead of trying 000000 to 999999
Try the first six digit of IC number

Try the default password
Try full IC number
Try student ID
Try studentID+IC
Try studentID+first 6 digit of IC

Try default username: admin
Try default passwords: admin, blank, admin123
Try default passwords from other brands

Username: use user email
Passwords: birthdate, girlfriends birthdate, {birthdate+gf birthdate}, favourite games name, Fengshui, 88888888…

JUST USE A WORDLISTS